INCH-POUND

MIL-L-52240C
17 March 1994
SUPERSEDING
MIL-L-52240B
9 April 1979

MILITARY SPECIFICATION

LANTERNS, ELECTRIC: LIGHT SOURCE, HEAD-MOUNTED, LOW-LEVEL, VISIBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers light-weight, self-contained, head-mounted lanterns designed to provide minimum illumination for use in areas where maximum security is required.

1.2 Classification. Lanterns covered by this specification shall be of the following types, as specified:

Type I - Hood, Unshielded

Type II - Hood, Shielded

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6230

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are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-V-173	-	Varnish, Moisture- and Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment)
MIL-T-704	-	Treatment and Painting of Materiel

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

TA13204E2450	-	Lantern, Electric; Light Source, Head-Mounted, Low-Level, Visible

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-UX, Natick, MA 01760-5017.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging
D 3953 - Strapping, Flat Steel and Seals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials and components. The materials and components shall be as specified herein and on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer, provided the completed item complies with all the provisions of this specification. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3 Design and construction. The design and construction shall be as specified herein and as shown on drawing TA13204E2450 and all subsidiary drawings pertaining thereto.

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3.3.1 Treatment and painting. The portions of the lantern normally painted shall be cleansed, treated, and painted in accordance with MIL-T-704, type A.

3.3.2 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.3.2.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed, need not be treated.

(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.3.2.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Southern California Air Pollution Control District, Rule 102 - Photochemically Reactive Solvents.

3.4 Identification marking. The lantern shall be identified in accordance with MIL-STD-130 and as shown on the applicable drawings.

3.5 Workmanship. Joints and seams shall be a tight fit, edges shall be smooth, and soldered surfaces shall be clean.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and shall be tested for the characteristics specified in 4.4.5. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Certification. The contractor shall furnish a certificate of compliance certifying that the varnish used to treat the electrical circuitry against fungus growth and moisture conforms to an is applied in accordance with the requirements specified in 3.3.2.

4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

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4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of lanterns. The sample unit shall be one lantern. The inspection level shall be II and the finding of any defect shall be cause for rejection of the lot.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Classification Minor
Painted surfaces	Blistered, crinkled, peeling, chipping, or not adherent		201
	Color not as specified		202
Workmanship	Incorrect fit at joints and seams, edges not smooth or any misalignment		203
	Soldering defects such as cold joints or excessive solder		204
	Any component missing, loose, fractured or malformed	101	
Design	Any nonconformance to the drawings	102	
Marking	Missing, incorrect, illegible, misplaced, or not as specified		205

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of lanterns. The sample unit shall be one lantern. The inspection level shall be S-2 and the finding of any defect shall be cause for rejection of the lot.

4.4.5 End item testing. The end items shall be tested for operation of the ON-OFF switch and removal and installation of the lighthood as specified in 4.5.1. The lot size shall be expressed in units of lanterns. The sample unit shall be one lantern. The inspection level shall be S-2. Any test failure shall be cause for rejection of the lot.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the finding of any defect shall be cause for rejection of the lot.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.5 Methods of inspection.

4.5.1 Operation. The lantern, with batteries installed, shall be operated by actuating the "ON" and "OFF" switch ten times and by installing and removing the lighthood (type II lantern) five times. Any malfunction of the lantern or failure of the lighthood (type II lantern) to lock into position when installed shall constitute failure of the test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Each lantern shall be individually preserved in accordance with MIL-P-116, method IC-2, using a close-fitting commercial fiberboard box or a box conforming to PPP-B-566 or PPP-B-636, type CF, class domestic, variety SW, grade 125. The contents shall be cushioned within the box in a manner to prevent movement or damage.

5.1.2 Level B preservation. Each lantern shall be individually preserved in accordance with MIL-P-116, method III, using a close-fitting box conforming to PPP-B-636, W6c, style optional. The contents shall be cushioned within the box in a manner to prevent movement or damage.

5.1.3 Commercial preservation. Lanterns shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. The lanterns, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. The gross weight of each box shall not exceed 200 pounds. Strapping shall conform to type I or IV size, as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

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5.2.2 Level B packing. The lanterns, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-636, V3c. The gross weight or size of each box shall not exceed the limitations of the box specification. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.3 Commercial packing. The lanterns, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The lanterns are intended for use in providing the minimum visible illumination required for performing special tasks and maintaining maximum security from detection.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Illumination
Lamp

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 99

Review activities:

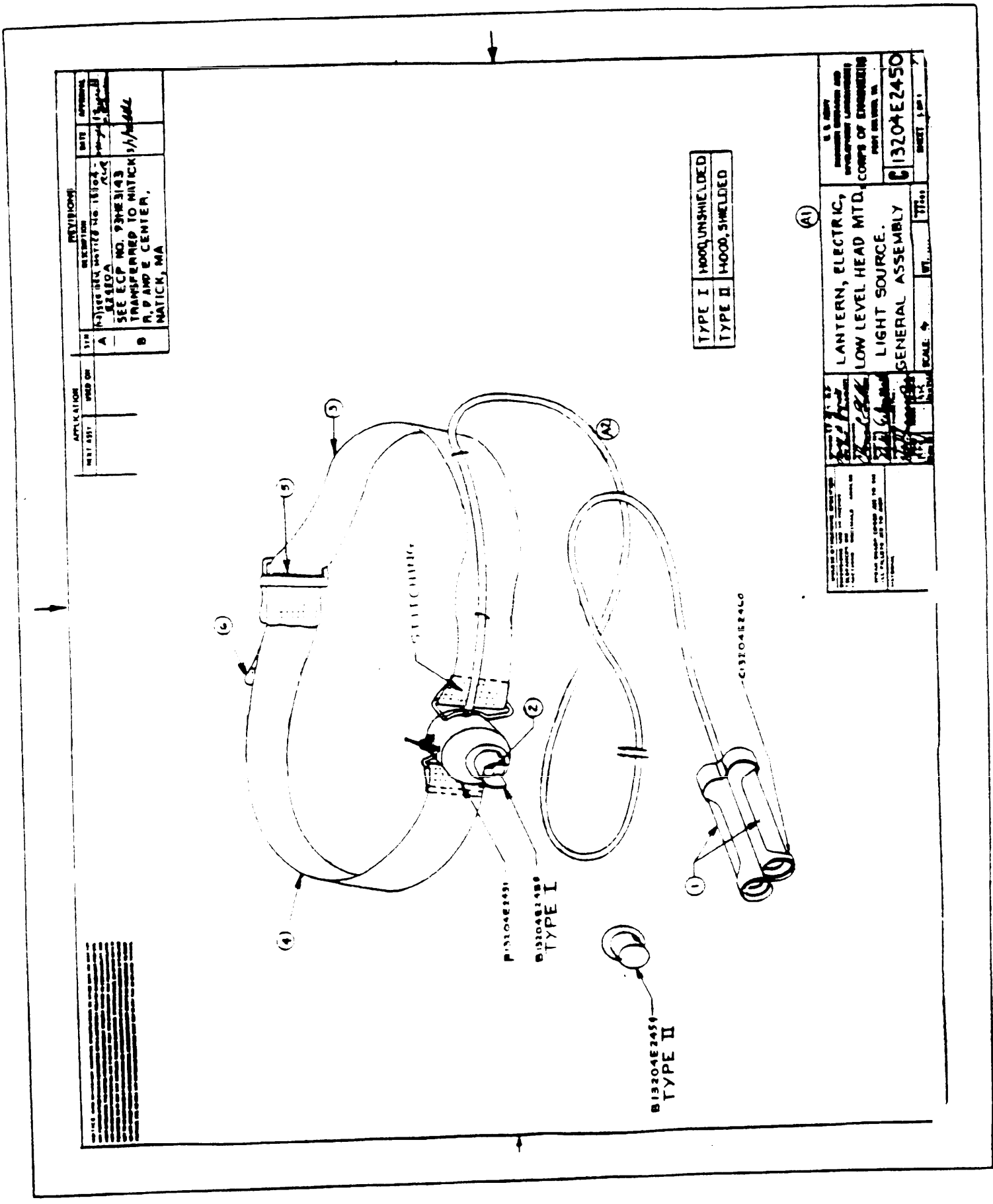
Air Force - 82
DLA - GS

Preparing activity:

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revisi letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-52240C	1994 March 17

3. DOCUMENT TITLE

LANTERNS, ELECTRIC: LIGHT SOURCE, HEAD-MOUNTED, LOW-LEVEL, VISIBLE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4532	(2) AUTOVON/DSN 256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-IR Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 NATICK OP2, JUL 91 *Previous editions are obsolete.*

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